Andre Arante

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Education

The George Washington University Washington, DC

B.S. Electrical and Computer Engineering, AI and Robotics Track *Aug 2023 – May 2027*

- Minor in Computer Science: coursework in Reinforcement Learning and Computer Vision; Technical GPA: 3.67
- Intro to Digital Signal Processing and Intro to Data Structures and Algorithms Teaching Assistant

Publications

1. **A. Arante**, M. Ghazanfari, C. Vu, and P. Wei, "Open-Vocabulary Obstruction Detection for UAS Landing via Fine-Tuned YOLO-World" accepted to *AIAA SciTech 2027*.

Work Experience

Robotics and AI Engineer Contractor Oct 2026 – Present

AIM Robotics, Inc *Herndon, VA*

- Working on robotic arm manipulation for pick-and-place tasks and perception systems for classifying 40,000 unique parts, trained in simulator environment Isaac Sim, and validated in lab and warehouse environments

Software Engineering Intern May 2026 – Aug 2026

Dell Technologies *Austin, TX*

- Fine-tuned VLA (SmolVLA) via imitation learning across multi-modal sensory inputs from a custom dataset of human demonstrations to perform autonomous hardware validation (keyboards) using LeRobot robotic arm
- Architected an RL post-training methodology that trains a small policy to output corrective actions improving task success from 40% to 90% and enabling the model to adapt to new keyboards with 2 hours of training time
- Accelerated policy training pipelines by 88% (66h to 8h) for arbitrary hardware validation tasks (e.g., port insertion) via efficient fine-tuning, sample-efficient transfer learning, and building a highly optimized demo recording pipeline

Undergraduate Research Intern Sep 2025 – Present

Intelligent Aerospace Systems Lab, The George Washington University *Washington, DC*

- Led research on fine-tuning open-vocabulary detectors (YOLO-World) for aerial-based obstruction detection
- Built an open-vocabulary benchmark (195 images, 8 unseen classes) to measure generalization collapse, then designed a selective layer-freezing strategy: 149% mAP gain while retaining 94% of baseline open-vocab confidence
- Built a VLM evaluation harness for obstruction detection and landing zone proposal on Jetson Thor/Orin/Xavier

Software Engineering Intern Feb 2024 – Feb 2025

C-3 Comm Systems, LLC *Bethesda, MD*

- Owned end-to-end development of an autonomous UGV platform, architecting a modular software stack that integrated computer vision, obstacle detection, and autonomous navigation which later became SortBots capstone
- Shipped and maintained in production an operations management system (JavaScript, Firebase) for 1+ year, automating payroll, time tracking, and logistics to replace a manual Google Sheets workflow

Projects

SortBots – Senior Capstone, Sponsored by C-3 Comm Systems

- Leading 5 engineers on a decentralized multi-robot package logistics system that adapts to unseen warehouses
- Integrated an RGB-D perception and navigation stack (Intel RealSense D435, Jetson Orin Nano, ROS 2) using RTAB-Map SLAM, 3D point cloud mapping, multi-robot map fusion, and Nav2 frontier exploration
- Built repeatable Isaac Sim test scenarios (e.g., workers crossing a robot's path) to evaluate obstacle avoidance
- Designed an electromagnetic end effector attachment in SolidWorks for reliable package pick-and-place

ImitationArm – Robotic Arm Teleoperation Device

- Developed a ROS 2 / C++ control stack integrating MoveIt2 for motion planning and MuJoCo for sim-to-real evaluation, teleoperated by a custom glove-based controller (ESP32 + MPU-6050) built for imitation learning

Technical Skills

ML & Robotics: PyTorch, CUDA, fine-tuning, imitation learning, motion planning, SLAM, vision-language-action models, reinforcement learning, Weights & Biases, NVIDIA Jetson, Linux, 3D Printing, CAD, Isaac Sim, LeRobot
Languages & Libraries: C++, Python, C, Matlab, C#, Java, ROS 2 (Nav2, MoveIt2), MuJoCo, OpenCV, SQL, AWS